



TRIOS CYBER

LABORATORY PRACTICE 8

Failed Authentication Event Triage and Disposition Analysis

Name: Matthew Josue Portilla Morejon

1. Previous Failed-Login Evidence Review

The authentication evidence generated during Laboratory Practice 7 was reviewed again from the Ubuntu endpoint. The journal records showed a remote SSH authentication sequence associated with the user matthew and source address 10.10.10.20. The sequence included a PAM authentication failure, three Failed password events, and a final connection-reset message.

```
matthew@matthew-VirtualBox:~$ sudo journalctl --since "2026-09-08 22:55:00" --until "2026-09-08 23:00:00" | grep -Ei "sshd|Failed password|authentication failure"
sep 08 22:56:58 matthew-VirtualBox sshd[5181]: Received disconnect from 10.10.10.20 port 49673:11: disconnected by user
sep 08 22:56:58 matthew-VirtualBox sshd[5181]: Disconnected from user matthew 10.10.10.20 port 49673
sep 08 22:56:58 matthew-VirtualBox sshd[5068]: pam_unix(sshd:session): session closed for user matthew
sep 08 22:57:29 matthew-VirtualBox sshd[5392]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=10.10.10.20 user=matthew
sep 08 22:57:31 matthew-VirtualBox sshd[5392]: Failed password for matthew from 10.10.10.20 port 49674 ssh2
sep 08 22:57:36 matthew-VirtualBox sshd[5392]: Failed password for matthew from 10.10.10.20 port 49674 ssh2
sep 08 22:57:41 matthew-VirtualBox sshd[5392]: Failed password for matthew from 10.10.10.20 port 49674 ssh2
sep 08 22:57:41 matthew-VirtualBox sshd[5392]: Connection reset by authenticating user matthew 10.10.10.20 port 49674 [preauth]
sep 08 22:57:41 matthew-VirtualBox sshd[5392]: PAM 2 more authentication failures; logname= uid=0 euid=0 tty=ssh ruser= rhost=10.10.10.20 user=matthew
```

2. Focused SSH Log Review

A focused review of the SSH service logs was performed for the original event window. The filtered output confirmed the same failed-login sequence and preserved the key fields required for triage: affected host, user, source address, source port, process, timestamps, and authentication result.

```
matthew@matthew-VirtualBox:~$ sudo journalctl -u ssh --since "2026-09-08 22:55:00" --until "2026-09-08 23:00:00"
-- Logs begin at Thu 2026-09-03 22:40:36 -05, end at Wed 2026-09-09 22:21:14 -05. --
sep 08 22:57:29 matthew-VirtualBox sshd[5392]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=10.10.10.20 user=matthew
sep 08 22:57:31 matthew-VirtualBox sshd[5392]: Failed password for matthew from 10.10.10.20 port 49674 ssh2
sep 08 22:57:36 matthew-VirtualBox sshd[5392]: Failed password for matthew from 10.10.10.20 port 49674 ssh2
sep 08 22:57:41 matthew-VirtualBox sshd[5392]: Failed password for matthew from 10.10.10.20 port 49674 ssh2
sep 08 22:57:41 matthew-VirtualBox sshd[5392]: Connection reset by authenticating user matthew 10.10.10.20 port 49674 [preauth]
sep 08 22:57:41 matthew-VirtualBox sshd[5392]: PAM 2 more authentication failures; logname= uid=0 euid=0 tty=ssh ruser= rhost=10.10.10.20 user=matthew
lines 1-7/7 (END)
```

3. Affected Asset and User Identification

The previous laboratory established the Ubuntu virtual machine as the destination of the remote authentication activity. The target system was matthew-VirtualBox at 10.10.10.10/24, with SSH listening on TCP port 22. The account involved in the failed authentication attempts was matthew, while the source was the authorized Windows laboratory endpoint at 10.10.10.20.

Triage Field	Observed Value	Interpretation
Affected asset	matthew-VirtualBox / 10.10.10.10	Ubuntu laboratory endpoint
Affected service	SSH / TCP 22	Remote authentication service
Affected user	matthew	Account targeted by the attempts
Source IP	10.10.10.20	Windows laboratory endpoint
Environment	VirtualBox Internal Network: LAB-NET	Isolated and authorized lab network

4. Timeframe and Event Sequence

The relevant remote-authentication activity occurred within a short interval on September 8, 2026. The first associated PAM authentication-failure record appeared at 22:57:29, followed by three Failed password records at 22:57:31, 22:57:36, and 22:57:41. The sequence ended with a connection reset and PAM summary at 22:57:41.

Time (-05)	Evidence	Triage Meaning
22:57:29	pam_unix(sshd:auth): authentication failure	Authentication failure begins
22:57:31	Failed password for matthew	Failed SSH attempt 1
22:57:36	Failed password for matthew	Failed SSH attempt 2
22:57:41	Failed password for matthew	Failed SSH attempt 3
22:57:41	Connection reset / PAM 2 more authentication failures	Failed sequence terminates

5. Likely Cause Assessment

The most likely cause was the controlled failed-login simulation performed during Laboratory Practice 7. The source IP, target account, destination service, attempt count, and timestamps match the known activity generated from the Windows endpoint to the Ubuntu SSH service inside LAB-NET. The evidence therefore supports an expected training event rather than unexplained external activity.

6. Severity Assessment

The event was classified as Low severity. Only three failed password attempts were generated, the source was a known laboratory endpoint, the destination was an isolated training system, and no successful authentication occurred during the failed-login sequence. The activity did not show evidence of account compromise or impact to systems outside the lab.

7. Triage Disposition

The recommended disposition is Close - Authorized Laboratory Activity. The evidence is fully explained by the previous controlled simulation, so escalation is not required. Similar authentication failures should be monitored or escalated when they originate from an unknown source, occur unexpectedly, increase substantially in volume, or are followed by successful unauthorized access.

8. Triage Summary

Field	Triage Result
Affected asset	Ubuntu endpoint matthew-VirtualBox (10.10.10.10)
Affected user	matthew
Source	Windows laboratory endpoint (10.10.10.20)
Timeframe	Sep 08 2026, 22:57:29-22:57:41 (-05)
Likely cause	Controlled SSH failed-login simulation from Laboratory Practice 7
Impact	No successful authentication during the failed-login sequence; no observed compromise
Severity	Low
Disposition	Close - Authorized Laboratory Activity

9. Conclusions

The failed-login evidence from Laboratory Practice 7 was successfully triaged using the Ubuntu SSH logs and the previously documented laboratory context. The affected asset, account, source endpoint, event timeframe, and authentication failure sequence were identified and correlated. Because the activity matched a known controlled test inside the isolated LAB-NET environment, it was assessed as Low severity and closed as authorized laboratory activity. This exercise demonstrated how SOC triage uses technical evidence and contextual information to determine whether an authentication event requires closure, continued monitoring, or escalation.